Weighted inequalities for maximal averages over manifolds

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What happened

- Stayed in Valencia working with David Beltran (He gave me an office!!)
- Spent three weeks in Budapest for a Summer School on the topic at the Rényi Institute. (And solved a problem with a friend! [BF26])
- Organized an internal seminar where the professor explained topics and we (a PhD student of his, and I) presented our results.
- Valencia is a great city but I also learned it is very hot

Averages over submanifolds and maximal operators

Let $S \subset \mathbb{R}^d$ be a compact submanifold with a finite positive measure μ, and

$$\mathcal{A}_t f(x) = \int_S f(x - ty)\, d\mu(y), \qquad Mf(x) = \sup_{t>0} |\mathcal{A}_t f(x)|.$$

- $S =$ ball: Hardy–Littlewood maximal function, bounded on L^p for all $p > 1$, by covering arguments.
- S of positive codimension: μ is singular, covering arguments fail, and boundedness depends on the curvature of S.

Theorem (Stein 1976, Bourgain 1986 [Ste76, Bou86])

The spherical maximal operator ($S = S^{d-1}$) is bounded on $L^p(\mathbb{R}^d)$ if and only if $p > \frac{d}{d-1}$.

- Threshold: $f(y) = |y|^{1-d}\big(\log \frac{1}{|y|}\big)^{-1} \mathbb{1}_{\{|y|<1/2\}} \in L^{d/(d-1)}$, yet $Mf \equiv +\infty$.
- Mechanism: focusing. $\mathcal{A}_t$ spreads a point over a sphere and, dually, concentrates a thickened sphere onto a small ball.

Power weights

Focusing happens at a point, so we measure it with a singular weight at one point:

$$\|f\|^p_{L^p(|x|^\alpha)} = \int_{\mathbb{R}^d} |f|^p \, |x|^\alpha \, dx. \qquad \text{For which } (p, \alpha) \text{ is } M \text{ bounded?}$$

Theorem (Duoandikoetxea and Vega 1996 [DV96])

For $p > \frac{d}{d-1}$ the spherical maximal operator is bounded on $L^p(|x|^\alpha)$ for

$$1 - d \; < \; \alpha \; < \; (d-1)p - d,$$

and this range is sharp (left endpoint: J. Lee [Lee25], $p \geq 2$).

Two examples explain the shape:

- *focusing*: $f =$ indicator of the δ-neighborhood of a sphere, so $Mf \approx 1$ on a ball B_δ: then $\delta^{d+\alpha} \lesssim \delta$ forces $\alpha \geq 1 - d$;
- *spreading*: $f = \mathbb{1}_{B_\delta}$, so $Mf \approx \delta^{d-1}$ on the unit annulus (take $t = |x|$): then $\delta^{(d-1)p} \lesssim \delta^{d+\alpha}$ forces $\alpha \leq (d-1)p - d$.

These are the only known obstructions; the difficulty is on the side of the proofs.

The Duoandikoetxea–Vega method: $\alpha \geq 0$

Only positivity is used. The key is a pointwise estimate for averages of the weight:

Lemma

$$\sup_{t>0} \mathcal{A}_t\big(|\cdot|^{-b}\big)(x) \;\leq\; C\,|x|^{-b} \qquad \textit{for} \quad 0 \leq b < d-1.$$

Proof: cap bound. The part of a sphere at distance s from the origin has measure $\lesssim s^{d-1}$, so the average of $|y|^{-b}$ over a sphere through the origin is $\lesssim \sum_i 2^{ib}\, 2^{-i(d-1)} < \infty$ exactly when $b < d-1$.

Fix $\frac{d}{d-1} < p_0 < p$. Hölder's inequality against the measure μ_t gives

$$\big(Mf\big)^p\,|x|^{\alpha} \;\leq\; C\big(MG\big)^{p_0}, \qquad \|G\|_{L^{p_0}}^{p_0} = \|f\|_{L^p(|x|^\alpha)}^{p},$$

provided $\frac{\alpha}{p-p_0} < d-1$. Integrating and using the unweighted theorem at p_0:

$$0 \;\leq\; \alpha \;<\; (d-1)(p-p_0) \;\xrightarrow[p_0 \to \frac{d}{d-1}]{}\; \alpha \;<\; (d-1)p - d \quad \text{(sharp)}.$$

The same family of estimates can be organized as a Stein–Weiss interpolation with change of measures [SW58], after linearizing the supremum. (Very useful!)

The Duoandikoetxea–Vega method: $\alpha < 0$

Here the weight rewards the focusing; four steps.

1. **Single scale.** Homogeneity reduces everything to $\|M_{\mathrm{loc}} f\|_{L^p(B_\delta)} \le C_\varepsilon\, \delta^{\frac{d-1}{p}-\varepsilon} \|f\|_{L^p}$, then dyadic sums give $\alpha > 1-d$.
2. **Space-localized estimate.** Cap bound at L^1, trivial L^2:

$$\|\mathcal{A}_t f\|_{L^1(B_\delta)} \le C\,\delta^{d-1}\|f\|_{L^1}, \qquad \|\mathcal{A}_t f\|_{L^2(B_\delta)} \le \|f\|_{L^2};$$

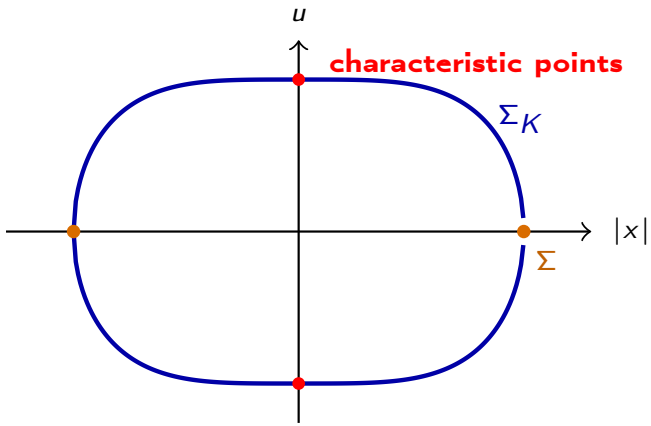
   Riesz–Thorin gives $\|\mathcal{A}_t f\|_{L^p(B_\delta)} \le C\delta^{(d-1)(\frac{2}{p}-1)}\|f\|_{L^p}$ for $1 \le p \le 2$.
3. **Sobolev in t.** $\sup_t |F|^p \le |F(1)|^p + p\int_1^2 |F|^{p-1}|F'|\,dt$ reduces the supremum to *space-time (local smoothing) norms*. The derivative costs 2^j per frequency piece; it is paid by the curvature, $|\widehat{\sigma}(\xi)| \le C(1+|\xi|)^{-\frac{d-1}{2}}$.
4. **Orthogonality in t.** Radial shells f_l of thickness δ reach $x \in B_\delta$ at disjoint times I_l, $|I_l| \approx \delta$:

$$\int_1^2 \|\mathcal{A}_t f\|^2_{L^2(B_\delta)}\,dt \;\le\; C\sum_l |I_l| \sup_{t\in I_l} \|\mathcal{A}_t f_l\|^2_{L^2(B_\delta)} \;\le\; C\,\delta^d\,\|f\|^2_{L^2}.$$

What we extended it to: the Heisenberg group

$\mathbb{H}^n = \mathbb{R}^{2n} \times \mathbb{R}$ with a noncommutative group law, dilations $\delta_r(x, u) = (rx, r^2 u)$, gauge $\rho = (|x|^4 + u^2)^{1/4}$, homogeneous dimension $Q = 2n + 2$. Two unit spheres:

- flat sphere $\Sigma = S^{2n-1} \times \{0\}$;
- Korányi sphere $\Sigma_K = \{\rho = 1\}$, with two characteristic points: gauge balls cut out measure $\approx r^{2n+1}$ at generic points, but only $\approx r^{2n}$ at characteristic!



Unweighted theory known [Cow81, MS04, NT04]; weighted theory did not exist before this work.

Why the direct Duo-Vega method is delicate on $\mathbb{H}^n$

The convolution is the group one, and the Euclidean Fourier transform of the spheres has no decay in the central direction. The frequency decomposition is much more delicate (the gauge is not linear!) and the analysis is finer, but we are working on it!

The blocker, and the $L^p \to L^q$ shortcut

The hard half of the method is the space-localized estimate: a gain $\delta^{c/p}$ for $\|M_{\mathrm{loc}} f\|_{L^p(B_\delta)}$.

The blocker

In the Duoandikoetxea–Vega proof the gain comes from Steps 3 and 4: frequency decomposition plus the decay of $\widehat{\sigma}$. Exactly the machinery missing on $\mathbb{H}^n$; the L^1/L^2 interpolation alone is far too weak.

The shortcut

Suppose the local maximal operator improves integrability: $\|M_{\mathrm{loc}} f\|_{L^q} \le C\|f\|_{L^p}$ with $q > p$. Then Hölder's inequality on the output ball gives the localized gain in one line:

$$\|M_{\mathrm{loc}} f\|_{L^p(B_\delta)} \;\le\; |B_\delta|^{\frac{1}{p}-\frac{1}{q}}\, \|M_{\mathrm{loc}} f\|_{L^q} \;\le\; C\,\delta^{Q(\frac{1}{p}-\frac{1}{q})}\, \|f\|_{L^p}.$$

All the oscillatory work is hidden inside the $L^p \to L^q$ estimate.

Known on Euclidean sphere [AHRS21], on $\mathbb{H}^n$ [RSS22, Sri24] and for the helix in $\mathbb{R}^3$ [BDH24]. For curves in $\mathbb{R}^n$, $n > 3$, the needed bound is conjectural.

A simpler scheme from $L^p \to L^q$ estimates

On $\mathbb{H}^n$ strong $L^p \to L^q$ estimates for $\sup_{1\le t\le 2} |\mathcal{A}_t|$ are known [RSS22, Sri24]. What do they give *alone*, with no frequency decomposition?

(H1) M bounded on $L^p(G)$ for $p > p_c$;

(H2) $\big\| \sup_{1\le t\le 2} |\mathcal{A}_t f| \big\|_{L^{q_1}} \le C \|f\|_{L^{p_1}}$ for $(1/p_1, 1/q_1)$ in a convex region $\mathcal{R}$;

(H3) $\sup_{t>0} \mathcal{A}_t \rho^\beta \le C \rho^\beta$ for $-\kappa_0 < \beta \le 0$ $(\kappa_0 = d-1$ for the sphere, $2n$ on $\mathbb{H}^n)$.

Proposition 1 (positivity; no linearization)

(H1)+(H3) $\Rightarrow$ M bounded on $L^p(\rho^\alpha)$ for $0 \le \alpha < \kappa_0(p - p_c)$.

Proposition 2 (three spatial regimes)

(H1)+(H2) $\Rightarrow$ bounded whenever $\varepsilon_1 = \frac{\alpha + Q(1 - p\, v_{\mathcal{R}}(1/p))}{p} > 0$ and $\varepsilon_2 = \frac{Q(p\, u_{\mathbf{2}} - 1) - \alpha}{p} > 0$.

Theorem

$-Q\Big(1 - p\, v_{\mathcal{R}}(1/p)\Big) \;<\; \alpha \;<\; \max\Big\{\kappa_0(p - p_c),\; Q\big(p\, u_{\mathcal{R}} - 1\big)\Big\}$, *plus Stein–Weiss convexification.*

$v_{\mathcal{R}}(u)$ = lower boundary of $\mathcal{R}$; $u_{\mathcal{R}}$ = its maximal abscissa.

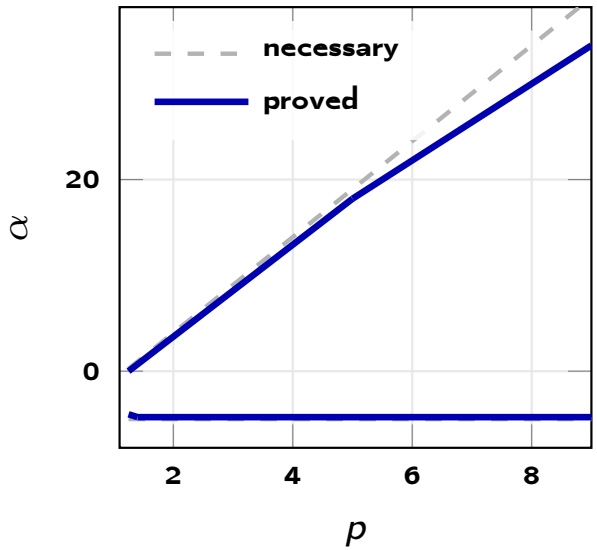
Results on $\mathbb{H}^n$, and what we learned

Evaluating the theorem on the regions of [RSS22, Sri24] gives, for both spheres, explicit piecewise linear boundaries $-A(p) < \alpha < B(p)$; at $n = 2$:

	left $-A(p)$	right $B(p)$
Korányi	$-(2p+2),\ -\frac{24}{5}$	$\frac{6}{5}(4p-5),\ 4p-2$
flat	$-6(3p-4),\ -6(p-1),\ -\frac{18}{5}$	$4p-\frac{16}{3}$

The gaps to the necessary conditions are small (left: $\frac{1}{2n+1}$, $\frac{2}{2n+1}$). Not sharp but close.

Korányi sphere, $n = 2$



What we learned

The $L^p \to L^q$ estimates already contain much of the information extracted by the frequency decomposition: when they are strong (large p), the output is sharp or close to sharp. Back-testing: the scheme recovers the full range of [DV96] for $p \geq \frac{d^2+1}{d(d-1)}$ (region of [AHRS21]), and the sharp helical range $-1 < \alpha < p-3$ of [GS26] for every $p \geq 4$ (region of [BDH24]). It is also sharp if we assume the conjectural estimates for curves. For small p, this analysis is much looser.

Gracias por su atención!









Gràcies per la vostra atenció!

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